Scene Adaptive SAR Incremental Target Detection via Context-Aware Attention and Gaussian-Box Similarity Metric

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***Abstract*— Existing incremental target detection (ITD) methods heavily depend on the diversity of information. When the scene in the new image diverges significantly from the previous training data, the detector's ability to detect known targets diminishes considerably. The scene information in synthetic aperture radar (SAR) images is closely linked to target types, as targets of the same class often emerge in similar environments. Consequently, various scenes are frequently introduced in conjunction with new classes, posing substantial challenges to the robustness of the SAR incremental target detector. To tackle this issue, this article proposes the context-aware attention (CAA) and the Gaussian-box similarity metric (GBSM) methods to enhance the scene adaptability of SAR incremental target detectors. First, the CAA operates during the feature knowledge transfer stage, which consists of a global relation module and a local attention (LA) module. It integrates the relationship between the target and its contextual information while preserving contextual awareness through knowledge transformation. Second, the GBSM establishes a constraint factor through both 2-D Gaussian modeling and distribution similarity measurement. It further modifies the incremental localization loss to reduce the impact of target-background contrast on the model's localization capability. We set up multiple data increment scenarios using the MSAR and SAR-Aircraft datasets. Comparative experimental results show that our method achieves better performance. In addition, the time consumption of training was recorded, and comparisons demonstrate that our method also offers advantages in efficiency.**

***Index Terms*— Context-aware attention (CAA), Gaussian-box similarity metric (GBSM), synthetic aperture radar (SAR) incremental target detection (ITD), target-scene correlation.**

I. Introduction

AS AN important application of synthetic aperture radar (SAR) imagery, SAR target detection plays an important role in civil and military fields. With the wide application of airborne and satellite-borne SAR imaging technology [1] in recent years, the available SAR images are more abundant, which also puts forward higher requirements on the performance and practicality of SAR target detection algorithms. Traditional SAR target detection methods rely on hand-designed features and classifiers, and the basic process can be divided into three stages: region selection, feature extraction, and classification [2], [3], [4], [5]. Among them, feature extraction is its core, and the representative algorithms are scale-invariant feature transform (SIFT) [6] and histogram of orientation gradients (HOG) [7]. Nonetheless, the performance of these algorithms in complex scenarios is not satisfactory enough, and the design of the feature extractor relies heavily on the experience of the researcher. In the data-intensive world of today, deep learning (DL) techniques are becoming more and more important, and the same is widely used in the field of SAR target detection [8], [9], [10], [11], [12], [13], [14], [15].

Compared to optical images, SAR images are more costly to acquire, so that obtaining a large number of richly categorized images at once is usually not feasible. With the continuous acquisition of new data from sensors, the model requires ongoing adaptation to new tasks. In the prevalent offline training mode, the model must update its parameters using all historical and new data. As the volume of data increases, this approach places significant pressure on storage and computational resources. Conversely, if only new data is used for updating, the model's ability to detect targets from previously learned classes significantly diminishes. This phenomenon is depicted in Fig. 1, where the model's primary focus shifts to achieving optimal performance on the new data due to the absence of information from the original dataset. With each iteration of parameter updates, new knowledge quickly overshadows the old knowledge, a problem known as "catastrophic forgetting" (CF) [16], [17], [18], [19].

Incremental learning (IL) clearly represents a more suitable approach for gradually increasing data. Unlike traditional methods, IL requires only new samples or a small subset of older samples to update the existing model. This approach effectively mitigates "CF" while avoiding the necessity for repeated training. Given the practicality of IL methods, many deep-learning-based target detection techniques have transitioned from offline learning to incremental paradigms.

IL was first applied to target classification, with notable methods including the replay-based iCaRL [18], [20], [21], [22], [23] and the regularization-based LwF [19], [24], [25].

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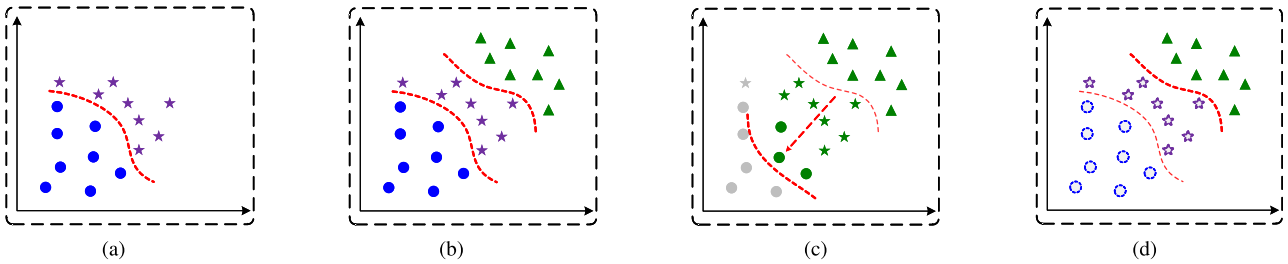


Fig. 1. Feature distributions in different cases are shown graphically. (a) Original model feature distribution, (b) desirable feature distribution after training by FD, (c) feature distribution in the case of CF, and (d) feature distribution after correction by IL method. (a) Original distribution. (b) Desirable distribution. (c) Memory conflict. (d) Knowledge revision.

Fast-ILOD [26] pioneered the integration of LwF into the field of target detection, using knowledge distillation (KD) to achieve incremental mechanism based on Fast-RCNN [27], but due to the nonend-to-end and poor practicality of this approach, subsequent researchers have gradually modified the end-to-end network. The representative algorithms are Faster-ILOD [28] based on the two-stage detector Faster-RCNN [29], ERD [30] based on the single-stage detector GFL [31], and SID [32] based on the anchor-free detector CenterNet [33]. All of the above methods belong to the IL methods based on regularization KD, in addition to meta-learning-based methods ONCE [34], replay-based methods Rodeo [35], and so on, but the KD-based methods are still the mainstream of incremental target detection (ITD).

However, due to insufficient consideration of the unique characteristics of SAR images, the aforementioned methods have not achieved satisfactory results in SAR target detection. The performance of incremental detectors is heavily influenced by the diversity of training data. When the scene information in new data significantly diverges from that of the prior, the robustness of the detector can deteriorate substantially, impairing its ability to detect previous targets [28], [36]. This issue is further compounded by the limited diversity in SAR images, where a strong correlation exists between targets and their background. Targets of the same class frequently appear in similar environments—ships are predominantly found at sea, while oilcans are typically located on land (as illustrated in Fig. 2). Consequently, the introduction of new target classes is frequently accompanied by the emergence of new scene contexts, requiring the detector to adapt dynamically to evolving scene conditions. This inherent correlation poses a significant challenge for ITD methods in SAR imagery.

Responding to the above issues, this study introduces the context-aware attention (CAA) and Gaussian-box similarity metric (GBSM) methods, with the aim of enhancing the adaptability and stability of SAR incremental target detectors across different scenes. Specifically, CAA is applied during the feature knowledge transfer phase of incremental training. This mechanism consists of parallel global relations and local attention (LA) branches, effectively capturing the contextual dependencies between targets and their environments. By implementing knowledge transfer, CAA not only strengthens the model's perception of contextual information but also effectively prevents the model from experiencing memory decay of historical scene knowledge.

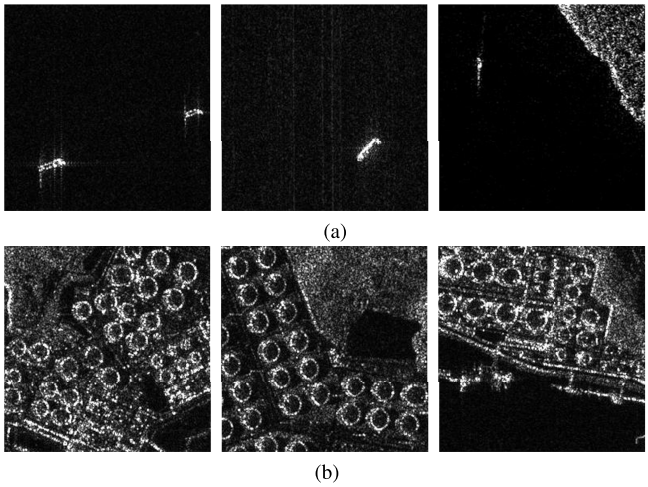


Fig. 2. Examples of SAR images. It can be seen that the ship and oilcan always appear in different scenes (sea and land), so there is a strong correlation between the target and background in the SAR image. But this unique property brings trouble to SAR ITD methods. (a) Ship. (b) Oilcan.

At the same time, the GBSM is used to reduce the over-sensitivity of incremental localization loss to the contrast between targets and backgrounds. The GBSM first models the target bounding box to 2-D Gaussian distribution. It subsequently generates constraint factors by evaluating the similarity between the distributions from the incremental model and the original model. This process effectively adjusts the alignment of the target localization output distributions between the incremental model and the initial model, encouraging the incremental detector to focus more on the central displacement and changes in the overall distribution, rather than solely relying on boundary information. Consequently, GBSM reduces the model's sensitivity to minor variations in target boundaries, significantly enhancing the adaptability and robustness of the incremental detector when facing complex scene changes.

The main contributions of this article are summarized as follows.

1) In this study, the shortcomings of previous ITD methods are analyzed according to the unique characteristics of SAR images. We deeply investigate the challenges posed by target-scene correlation in SAR images that affect the robustness of incremental object detectors.

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2) To enhance the model's perception of the potential relationships between targets and scenes, this article designs the CAA method. The CAA operates during the feature knowledge transfer phase of IL. By employing the knowledge transfer, it effectively integrates the relationship information between targets and their contextual information into the IL process. This improves the model's ability to perceive environmental relationships and retain existing knowledge.
3) To reduce the impact of scene changes on the localization capabilities of the incremental detector, this article proposes the GBSM method. This method corrects the incremental localization loss by fitting Gaussian distributions to the target bounding boxes and assessing the similarity between these distributions. This method reduces the sensitivity of the detector to the target-background contrast, thus enhancing the adaptability and stability of the detector in different scenes.

The subsequent sections are outlined as follows. In Section II, we review the common literature related to IL and ITD. In Section III, we present a detailed description of the innovative components in our proposed method. In Section IV, we validate the effectiveness of our method on the multiclass SAR target detection datasets MSAR-1.0 [37] and SAR-AIRcraft [38]. Finally, we summarize and discuss the method proposed in this article.

II. Related Work

CF is the main challenge of IL methods. Nowadays, there have been numerous works working on this problem, mainly focusing on the field of target classification. Comparatively, there are fewer IL methods in the field of target detection.

A. Incremental Learning

Mainstream IL methods can be categorized into two approaches based on the utilization of old samples: replay-based methods and regularization-based methods.

The replay-based IL method uses partially representative samples to convey knowledge from previous tasks. The iCaRL [18] is the most classical replay-based algorithm. After each training session, the image closest to the center of the feature mean is selected as a representative sample for subsequent training. Improved algorithms based on iCaRL, such as [20], [21], [22], and [23], enhance it through refined sample selection strategies. In addition to this, there are also some works [39] that introduce the GAN [40] network into IL by generating old samples through the GAN network to participate in the IL process. Replay-based IL methods have a wide application in the field of SAR target recognition, and representative works include [41], [42], [43], which focus on representative sample selection ways that are more applicable to SAR images.

The core principle of regularization-based IL methods is to prevent the overwriting of old knowledge with new information. Its typically achieved by imposing constraints through KD and continuously activating memory nodes associated with prior knowledge. LwF [19] is the most representative regularized IL method and the first to apply KD to IL. Through KD, the incremental model is forced to mimic the output of the original model in order to achieve the purpose of constraint. This idea has been followed by numerous subsequent works, representative works include low-dimensional feature mapping algorithm [24] and Bayesian framework [25]. Similarly, numerous regularization-based IL methods [44], [45], [46] have been applied to SAR target recognition in recent years. The most recent work [46] designed a multilevel adaptive KD network, which achieves desirable results in SAR target recognition.

B. Incremental Target Detection

ITD is the focus of this article. Compared to incremental target recognition, the ITD method is more challenging and at the same time has less related work. Fast-ILOD [26] is the first work to introduce IL methods to deep target detection by adapting the LwF algorithm for Fast-RCNN [27]. It is noteworthy that this is not an end-to-end detection method, as the locator of Fast-RCNN uses a separate selective-search algorithm and does not share parameters with the classifier, which can be more easily adapted, but also consequently makes the method lacking in practicality. Subsequent researchers have gradually turned their attention to end-to-end networks. Faster-ILOD [28] applies LwF to the two-stage detector Faster RCNN [29] by exploring the robustness of RPN. SID [32] achieves IL through KD and parameter isolation based on the anchor-free detector CenterNet [33]. ERD [30] applies the IL method to the single-stage detector GFL [31] using localization distillation [47] and EWC [25]. MMA [48] specifically addresses the problem of annotation missing through unbiased classification loss and unbiased distillation loss. Recent work IDCOD [49], improves the detection performance of incremental detectors for remote sensing targets by instance distillation and pseudo-labeling techniques. In addition to regularization-based methods, there are also ITD implemented by other means. Such as, ONCE [34] based on meta-learning, Redo [35] based on feature vector replay, and ABR [50] combining KD and instance-replay. Overall, ITD methods based on KD are still mainstream, capable of achieving incremental model updates without using previous data at all.

III. Method

A. Preliminary

1) Problem Definition: Let M_0 be the detection model trained with dataset C_0, and there exists a new dataset C_1, with $C_0 \cap C_1 = \varnothing$. Our task is to update the M_0 using only C_1 to obtain a new detection model M_1. M_1 is able to detect targets of the classes contained in C_1, while maintaining the ability to detect C_0.

2) Basic ITD Framework: ITD is a method that can iteratively update the existing model with new data to adapt to new target detection requirements. When images containing new classes are received, the new data is employed for incremental training, guided by existing knowledge. The whole process is shown in Fig. 3, where new images are fed into both the

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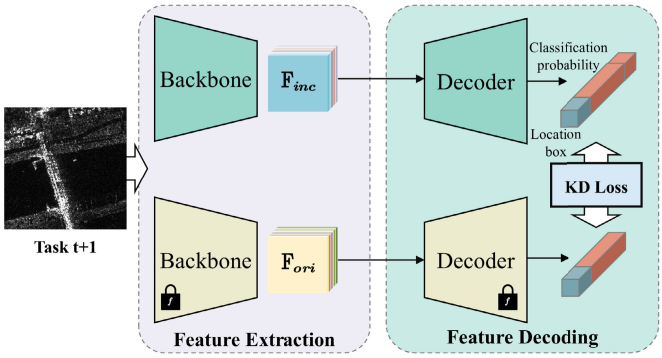


Fig. 3. Training process for ITD.

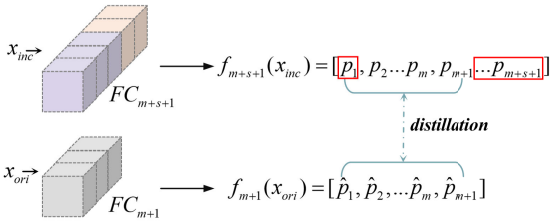


Fig. 4. IL process of classifier. x represents the features from the backbone which is used as input to the classifier. p is the probability distribution of the classifier output, and p_i is the confidence value for the ith class.

existing model (called the original model, M_{ori}) and the new model (called the incremental model, M_{inc}). M_{inc} acquires new knowledge from the incoming data while safeguarding prior knowledge via the output from M_{ori}. During feature extraction, both models output their feature maps and transfer them to their respective decoders. Notably, for M_{inc} and M_{ori} to work on the same objects, M_{ori} must rely on the proposals generated by M_{inc}. M_{ori} does not need to generate accurate results for M_{inc} but only provides its own judgments, which serve as "pseudo-labels."

The decoder determines the class and location using the classifier and regressor. Likewise, incremental detection methods necessitate transforming both the classifier and regressor.

For the classifier, the initial step in incremental training involves expanding the original branch. Assuming the original dataset includes class m and the new dataset contains class s, the classification branch will be expanded from $m+1$ to $m+s+1$ layers (with an additional layer for the background). During the training process, the classifier of the M_{inc} performs new class training with the first $m+1$ layers fine-tuned by the output of the original classifier. The whole process is shown in Fig. 4.

For the regressor, previous works treat it as a class-independent component, making only minor modifications, such as directly freezing the regressor parameters or performing "hard-distillation" of the results [28], [32]. This practice heavily tests the generalization ability of the initial model, and as a result, the model's localization performance significantly decreases after multiple increments. GFL [31] introduced a general representation of the regression boundary, which enabled the regressor output to be transformed into a probability distribution, similar to the classification. This approach was soon applied to KD [47] and ITD methods [30]. In this work, GFL is used as the base detector. The boundary position $b = \{t, b, l, r\}$ are converted into their corresponding distributions via the following formula:

$$\begin{aligned} &y = [y_0, y_1, \ldots, y_k] \\ &\Delta_i = y_i - y_{i-1}, \ |\Delta_i| = \left|\Delta_j\right| \quad \forall i, j \in \{1, 2, 3, \ldots, k\} \quad (1) \\ &y_i^* = \sum_{i=0}^{k} \mathcal{S}(\text{pred}(i)) \cdot i \\ &\sum_{i=0}^{k} \mathcal{S}(\text{pred}(i)) \cdot \Delta = 1. \quad (2) \end{aligned}$$

Through the general representation of the boundary, the space is first dispersed. Then, the localization confidence on each interval is calculated and normalized to get the probability distribution of each boundary localization, calculated as shown in (2). The transformed boundary is denoted as $b_t = [p_t, p_b, p_l, p_r]$, and the loss of the M_{inc} versus the M_{ori} is subsequently computed by KL scatter.

The training loss of the basic framework for ITD can be represented by (3), where $\mathcal{L}_{\text{gen}}$ represents the general learning loss, and $\mathcal{L}_{\text{cls}}^{\text{inc}}$ and $\mathcal{L}_{\text{reg}}^{\text{inc}}$ represent the classification and localization loss in knowledge review. Our method is based on the above-mentioned basic framework, which is described in detail in Section III-E

$$\mathcal{L}_{\text{basic}} = \mathcal{L}_{\text{gen}} + \lambda\left(\mathcal{L}_{\text{cls}}^{\text{inc}} + \mathcal{L}_{\text{reg}}^{\text{inc}}\right). \quad (3)$$

B. Overall Framework

Using the approach in the previous section as a basic framework, we implemented our proposed method, and the overall framework is shown in Fig. 5. The core of our method is the CAA with GBSM, which is used to correspond to the problems posed by the target-scene relevance specific to SAR images during the incremental process.

The CAA comprises two branches: global relation (GR) and LA, both of which play a role in the process of feature knowledge transfer. While GR extracts the long-range dependencies between the target and the background, LA focuses on capturing the LA of the target. Specifically, after the feature maps are output by the original model M_{ori} and the incremental model M_{inc}, these feature maps are, respectively, transformed into global relation maps and LA maps. This transformation aims to capture and highlight critical contextual information between targets and scenes within the feature map. Ultimately, the differences in information loss calculated by the GR and LA branches are integrated into the knowledge review process. CAA not only facilitates a deeper understanding of key contextual information by the model but also ensures that the model does not forget existing important information while learning new knowledge.

The GBSM is utilized to optimize incremental localization loss, with the goal of reducing the impact of target-scene contrast on the definition of bounding box boundaries, thereby enhancing the detector's adaptability to various scene changes. To this end, we employ 2-D Gaussian distributions to model

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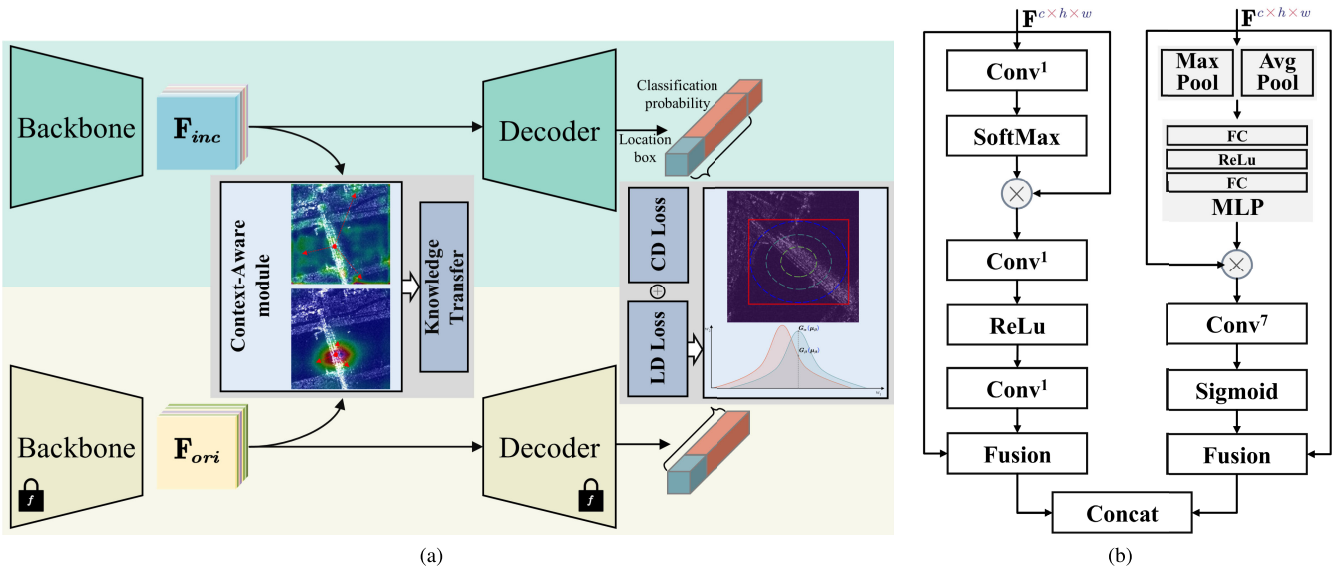


Fig. 5. Overall framework diagram of our method depicts the IL process divided into three phases. In the first stage, feature extraction, SAR images are fed into both the incremental model and the original model, and it is worth noting that the parameters of the original model are in a frozen state. In the second stage, feature knowledge transfer, the CAA processes the feature outputs of the two models to obtain the corresponding CA maps, and compares the differences to obtain the context-aware loss $\mathcal{L}_{\text{ca}}$. In the third stage, feature decoding, after the detector head of the two models outputs the classification and regression scores, KD is carried out to obtain the incremental loss $\mathcal{L}_{\text{inc}}$, which contains the incremental classification loss $\mathcal{L}_{\text{cls}}^{\text{inc}}$ and the incremental localization loss $\mathcal{L}_{\text{reg}}^{\text{inc}}$. Meanwhile, based on the target location and size output from the two models, a GBSM is established to constrain the $\mathcal{L}_{\text{reg}}^{\text{inc}}$, denoted as $\mathcal{L}_{G}^{\text{inc}}$. (a) Overall Framework. (b) Structure of the CAA.

the positions of the regression boxes. In this process, the height and width of the target are used to determine the standard deviation of the Gaussian distribution, while the center of the target serves as the mean. Given that changes in the standard deviation directly influence the Gaussian density at the center, by comparing the Gaussian distribution models of the localization results for the same target from M_{ori} and M_{inc}, it is possible to accurately calculate the degree of central displacement and effectively measure the differences in the overall distribution. This method not only reduces the sensitivity of bounding box boundary definitions to background interference but also enhances the positioning accuracy and robustness of the incremental detector in diverse scenes.

The implementation of the CAA and GBSM will be detailed in Sections III-C and III-D.

C. Context-Aware Attention

In SAR imagery, targets belonging to the same class typically manifest within similar scenes. Consequently, the introduction of new target classes is invariably associated with the emergence of new scenes. Prior research [28] has demonstrated that changes in the scene context throughout the IL process can significantly impair the detector's performance in localizing previously known targets. Therefore, it is essential to establish and maintain a robust category-to-scene relationship during the incremental detection of SAR targets.

According to the above ideas, this article proposed the CAA as shown in Fig. 5(b). Inspired by works [51], [52], and [53], this method combined the GR with LA, and applied it to the process of knowledge review.

1) GR: The structure of the GR module is shown in the purple part of Fig. 5(b), the structure of the GR module is illustrated in the left branch of Fig. 5(b), and its core idea is to calculate the pair-wise relationship between each position of the feature map and all the positions, form the correlation matrix, and finally fusion with the feature map. The module can be abstracted into three parts: global feature extraction, feature transformation, and feature fusion.

The role of global feature extraction is to capture the correlation between a feature point and the global feature, which is the core of the GR module. For a feature map $\mathbf{F}$ with dimension $C \times W \times H$, a compressed feature map with dimension $1 \times W \times H$ is obtained after compression of the 1×1 convolutional layer w_1, denoted as $x = \{x_i\}_{i=1}^{N}$. For each feature point x_i of the compressed feature map, the correlation factor between x_i and each position is obtained by calculating the attention weight of x_i with respect to the global feature. Iterate over all positions of the feature map and compute as follows:

$$x_{v1} = \sum_{j=1}^{N} \left(\frac{e^{w_1 * x_j}}{\sum_{m=1}^{N} e^{w_1 * x_m}} \cdot x_j \right) \tag{4}$$

where x_j denotes the location of all possible enumerations. After the correlation is computed for each feature point of the compressed feature map, a correlation sequence of dimension $C \times 1 \times 1$ is obtained, denoted as x_{v1}.

The role of feature transformation is to normalize the correlation sequence while building interdependencies between channels through dimension expansion. Previous research [53] has shown that weighting feature channels by contextual relationships can improve detector performance. The feature transformation part consists of convolutional layer w_2, normalization layer BN, and convolutional layer w_3, with input x_{v1} to obtain the weight matrix in dimension $C \times 1 \times 1$, which

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is calculated as follows:

$$x_{v2} = w_3 * \mathrm{BN}(w_2 * x_{v1}). \tag{5}$$

The contextual features are fused to the features at all locations, i.e., the feature map $\mathbf{F}$ is channel-weighted using a weight matrix to obtain the final GR map. The whole process is computed as follows:

$$\mathbf{F}^{\mathrm{G}} = \mathbf{F} + w_3 * \mathrm{BN}\left(w_2 * \sum_{j=1}^{N}\left(\frac{e^{w_1 * x_j}}{\sum_{m=1}^{N} e^{w_1 * x_m}} \cdot x_j\right)\right). \tag{6}$$

2) Local Attention: Inspired by CBAM [54] and SeNet [53], the structure of the LA module is shown in the right branch of Fig. 5(b), and the core idea is to merge the attention after acquiring it in both the channel and spatial directions, and ultimately fusing it with the feature map. The module can be abstracted into three parts, channel attention, spatial attention, and attention fusion.

Channel attention enables features to be corrected to retain more valuable ones, allowing the network to pay less attention to channels without important information. The first step of channel attention is to compress the original input features in the spatial dimension to obtain a 2-D vector for subsequent operations, which is achieved by global average pooling (GAP) and global maximum pooling (GMP). GAP has feedback for every feature point on the feature map, while GMP has feedback only for the feature points with the largest response in the backpropagation, the combination of both together accomplishes attention extraction and feature dimensionality reduction. The process is computed as follows:

$$\begin{aligned} y_a^c &= \{y_i\}_{i=1}^{C} \; y_i = \frac{1}{|\mathcal{R}_i|}\sum_{(p,q)\in\mathcal{R}_i} x_{pq} \\ y_m^c &= \left\{y_j\right\}_{j=1}^{C} \; y_j = \max_{(p,q)\in\mathcal{R}_j}\left(x_{pq}\right) \\ y^c &= y_a^c \oplus y_m^c \end{aligned} \tag{7}$$

where y_a^c and y_m^c represent the 1-D vectors obtained after GAP and GMP, $\mathcal{R}$ is the feature region on each channel, and x_{pq} is the eigenvalue located at (p, q) within the region. For the dimension $C \times W \times H$ feature map $\mathbf{F}$, the 2-D vector is finally obtained after the above operations. In the next step, a vector of weights is generated by Shared-MLP on the y^c network connection, which is multiplied with the original feature map to obtain the calibrated attention feature map $\mathbf{F}^{\mathrm{c}}$. The whole process operates as follows:

$$\mathbf{F}^{\mathrm{c}} = \sigma\left(f_{\mathrm{mlp}}\left(y_a^c\right) + f_{\mathrm{mlp}}\left(y_m^c\right)\right) \times \mathbf{F}. \tag{8}$$

Spatial attention complements channel attention by finding the densest regions of information distribution based on the direction of channel attention. The first step of spatial attention extraction is still implemented through GAP and GMP, unlike channel attention, the pooling operation of spatial attention is performed along the channel dimension. The process is computed as follows:

$$\begin{aligned} y_a^s &= \{y_i\}_{i=1}^{W\times H} \; y_i = \frac{1}{|\mathcal{C}_i|}\sum_{m\in\mathcal{C}_i} y_m \\ y_m^s &= \left\{y_j\right\}_{j=1}^{W\times H} \; y_j = \max_{m\in\mathcal{C}_j}(y_m) \\ y^s &= y_a^s \oplus y_m^s \end{aligned} \tag{9}$$

where y_a^s and y_m^s represent the 2-D feature matrix obtained after GAP and GMP, $\mathcal{C}$ is the feature sequence at each spatial position, and y_m is the mth eigenvalue of the sequence. For the input feature map $\mathbf{F}^{\mathrm{c}}$, the feature descriptor y^s with dimension $2 \times W \times H$ is obtained after the above operation. Spatial filtering and activation operations are performed on y^s and fused with the original input feature map $\mathbf{F}$ to obtain the final LA map. The scale of spatial filtering is set to 7×7 according to the suggestion in work [53], and the activation function is $sigmoid$, which can calculated as follows:

$$\mathbf{F}^{\mathrm{L}} = \sigma\left(w^{7\times 7} * \mathbf{F}^{\mathrm{c}}\right) \times \mathbf{F}. \tag{10}$$

3) Incremental Context-Aware Loss: In order to use CAA to guide model training, it is necessary to form the loss to be added to the IL process after obtaining contextual information. The computation of incremental context-aware loss ($\mathcal{L}_{\mathrm{ICA}}$) is similar to feature-based KD [55], except that we do not directly compare the differences between features. The formulas are shown in (11), where $\mathbf{F}^{\mathrm{G}\alpha}$ and $\mathbf{F}^{\mathrm{G}\beta}$ are the global relational feature maps for the original and the incremental model, and $\mathbf{F}^{\mathrm{L}\alpha}$ and $\mathbf{F}^{\mathrm{L}\beta}$ are the LA maps of the original and the incremental model, using L2 loss to compute the difference between each position

$$\begin{aligned} \mathcal{L}_{\mathrm{ICA}} = &\frac{1}{CWH}\sum_{k=1}^{C}\sum_{i=1}^{W}\sum_{j=1}^{H}\left(\mathbf{F}_{k,i,j}^{\mathrm{G}\alpha} - \mathbf{F}_{k,i,j}^{\mathrm{G}\beta}\right)^2 \\ &+ \frac{1}{CWH}\sum_{k=1}^{C}\sum_{i=1}^{W}\sum_{j=1}^{H}\left(\mathbf{F}_{k,i,j}^{\mathrm{L}\alpha} - \mathbf{F}_{k,i,j}^{\mathrm{L}\beta}\right)^2. \end{aligned} \tag{11}$$

D. Gaussian-Box Similarity Metric

Since the regression branch does not expand with the increase in the number of categories, the localization capability of the incremental detector exhibits higher sensitivity to the match of knowledge domains. This means that when there is a significant difference between the scene information in the new dataset and that of the previously trained data, the model's performance tends to decline markedly. A key reason for this phenomenon lies in the fact that the detector relies on the boundary fitting of bounding boxes to determine the specific location of targets, and the effectiveness of boundary fitting is highly susceptible to the contrast between targets and backgrounds. Therefore, when there are significant changes in background information, the model's localization accuracy may be severely impacted. To address this issue, this article proposes the GBSM method, which aims to adjust the incremental localization loss by quantifying the similarity between Gaussian distributions. This approach effectively reduces the model's sensitivity to the boundaries of bounding boxes.

The 2-D Gaussian distribution is nonuniform, and the density value of each position (Gaussian distribution density function, pdf) is affected by the target size and the distance between the position and the center point at the same time. The target size will make its distribution dispersion, thus

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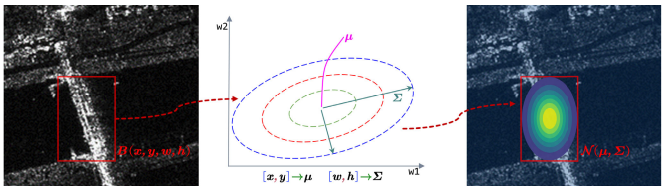


Fig. 6. Schematic of the 2-D Gaussian distribution transformation to create a Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ based on the target center coordinate (x, y) and size (w, h).

reducing the density, otherwise higher density; and the closer the distance from the center point the higher density, and vice versa, the density is reduced. Therefore, 2-D Gaussian modeling based on the positioning of the target rectangle box and comparing the difference in the positioning distribution between the original model and the incremental model is able to measure the degree of positioning and center point offset of both at the same time. The modeling process is shown in Fig. 6, for the target X, the general bounding box representation marks its position as $B(x, y, w, h)$, (x, y) as the center position, and (w, h) as the width and height of the target. Based on the target position and dimension information, the mean μ and variance Σ of the Gaussian distribution are calculated as follows:

$$\mu = [x, y]^T = [\mu_1, \mu_2]^T \tag{12}$$

$$\Sigma = \begin{bmatrix} \left(\frac{w}{2}\right)^2 & 0 \\ 0 & \left(\frac{h}{2}\right)^2 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \\ & \sigma_1^2 \end{bmatrix}. \tag{13}$$

The transformation relationship between the 2-D Gaussian distribution and the rectangular box can obtained through the mean μ and variance Σ. General 2-D Gaussian distribution density is shown in (14), where ρ is the associated parameter of μ_1 and μ_2

$$\begin{aligned} &\mathcal{N}(\mu, \Sigma) \\ &= \left(2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}\right)^{-1} \\ &\quad \times \exp\left(-\frac{1}{2}\left(\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right)\right). \end{aligned} \tag{14}$$

Considering that the center coordinates x and y are independent of each other hence the ρ-value is 0, then the calculation process is as follows:

$$\mathcal{N}(x, y; \Sigma) = \frac{2}{\pi wh}\exp\left(-2\left(\frac{(x\prime - x)^2}{w^2} + \frac{(y\prime - x)^2}{h^2}\right)\right). \tag{15}$$

In this case, the Gaussian distribution model of the target X with coordinate $B(x, y, w, h)$ can be written as $X \backsim \mathcal{N}(x, y; \Sigma)$.

Furthermore, the localization outputs of M_{ori} and M_{inc} are modeled with Gaussian distribution, and the differences between the two distributions are compared. Let the localization prediction of M_{ori} and M_{inc} for the target X be B_α and

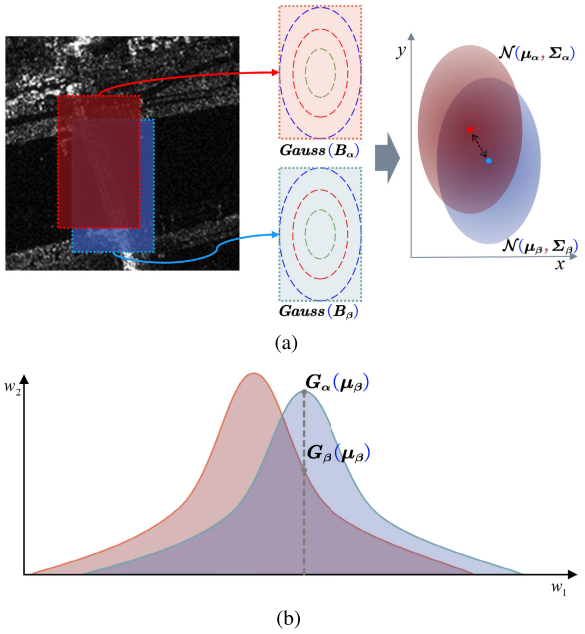


Fig. 7. Diagram of GBSM calculation. It is first converted to a Gaussian distribution based on the predicted locations and sizes, and then the constraint factor is formulated by measuring the difference between the two distributions. (a) 2-D Gaussian distribution transformation. (b) Similarity metric.

B_β, respectively, and first perform the Gaussian distribution transformation as shown in Fig. 7(a). The transformation result is denoted as $\mathcal{N}(\mu_\alpha, \Sigma_\alpha)$ with $\mathcal{N}(\mu_\beta, \Sigma_\beta)$. As mentioned before, the center density of Gaussian distribution is affected by the size of the target, so in this article, we measure the degree of localization bias by the difference in the density of the centroid position μ_β predicted by the incremental model in two Gaussian distributions. The idea is illustrated in Fig. 7(b). The smaller difference between the two represents a higher degree of distribution similarity. At the same time, the measurement is constrained by the target scale, for example, when the centroids of the two distributions coincide, the center densities are likewise different if the scales are different. The above method is able to take into account the difference in scale while measuring the degree of center offset, and greatly reduces the amount of computation compared to directly calculating the complete difference between the two distributions.

Finally, there is the construction of the constraint function. The most intuitive way to do this is to directly compute the difference between the two and normalize it, which is similar to IoU, but obviously not reasonable for Gaussian. Since the Gaussian is a form of diffusion from the center to the edges, there is a large gradient of density decrease near the center of the distribution, which can lead to a function that is unusually sensitive in the center region. For this reason, in order to obtain more smoother constraint function, we constructed the following three constraint functions by borrowing from the mainstream accuracy evaluation methods.

1) Intersection over union (IoU) like

$$\frac{\left|G_\alpha(\mu_\beta) - G_\alpha(\mu_\alpha)\right|}{G_\alpha(\mu_\alpha)}. \tag{16}$$

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2) F1 like

$$\frac{\sqrt{G_\alpha(\mu_\beta)G_\beta(\mu_\beta)}}{2\left(G_\alpha(\mu_\beta)+G_\beta(\mu_\beta)\right)}. \tag{17}$$

3) Wasserstein like

$$\left\|\mu_\alpha-\mu_\beta\right\|_2^2+\left\|\Sigma_\alpha^{1/2}-\Sigma_\beta^{1/2}\right\|_{\text{Frobenius}^*}^2. \tag{18}$$

The disadvantage of the first approach is that the constraint is not smooth enough, resulting in less-than-desirable results. The second approach is a relatively optimal way to ensure smoothness while greatly ensuring computational efficiency. The last approach is theoretically the most accurate, but the computational effort is much greater than the former, and there is no additional focus on the center of attention information in this approach. Therefore, in this article, the second construction method will be taken. The corrected incremental localization loss is denoted as $\mathcal{L}_G^{\text{inc}}$, and its computation process is as follows:

$$\mathcal{L}_G^{\text{inc}}=\frac{\sqrt{G_\alpha(\mu_\beta)G_\beta(\mu_\beta)}}{2\left(G_\alpha(\mu_\beta)+\right)G_\beta(\mu_\beta)}\sum_B(p\log p-p\log q). \tag{19}$$

E. Total Loss

In summary, the overall loss function consists of three components. The general learning loss $\mathcal{L}_{\text{gen}}$, which represents the loss of the incremental model to learn new classes of knowledge, and the learning process is exactly the same as the usual off-line training way. The incremental loss $\mathcal{L}_{\text{inc}}$, which represents the knowledge inherited from the original model by means of KD, is in turn composed of two parts: the incremental classification loss $\mathcal{L}_{\text{cls}}^{\text{inc}}$ and the incremental localization loss $\mathcal{L}_G^{\text{inc}}$. The incremental context-aware loss $\mathcal{L}_{\text{ICA}}$, is the loss proposed in this article for capturing the context relationship of SAR targets during the IL process. The overall loss function is as follows:

$$\mathcal{L}=\mathcal{L}_{\text{gen}}+\lambda_1\left(\mathcal{L}_{\text{cls}}^{\text{inc}}+\mathcal{L}_G^{\text{inc}}\right)+\lambda_2\mathcal{L}_{\text{ICA}}. \tag{20}$$

The λ_1 and λ_2 in the loss $\mathcal{L}$ are balancing parameters to control the participation of $\mathcal{L}_{\text{inc}}$ and $\mathcal{L}_{\text{ICA}}$ in the whole IL process. For the setting of the parameters, this article is discussed in detail in Section IV.

IV. Experiment

In this section, we validate the effectiveness of our method using two SAR target detection datasets: MSAR [37] and SAR-AIRcraft [38]. At the beginning, we provide an introduction to both datasets and explain the process of preprocessing, as well as the evaluation metrics used in experiments. In the section of experimental settings, different training strategies and IL scenarios during the class-incremental process are described in detail, further based on different scenarios, we partition the datasets. In the ablation study, we observe the performance of our method under different parameters and verify the effectiveness of each component. Finally, we compare the performance of our method with two state-of-the-art one-stage ITD methods in different scenarios. All experiments in this section were run three times with the same setting, and the best of three runs was taken as the final result.

All methods in this chapter are implemented on the MMDetection platform and the same settings are used for the training process. The experiments were all done on NVIDIA RTX 4090 and Ubuntu 20.04 devices.

A. Datasets and Evaluation Metrics

MSAR-1.0 [37] uses the data from Hisea-1 and Gaofen-3 satellites, with polarization modes including HH, HV, VH, and VV. The dataset contains a total of 28 449 images and 60 456 targets, and the image size is 256 × 256 pixels (some bridge images are 2048 × 2048 pixels). The scenes in the MSAR dataset are relatively plentiful, including airports, harbors, offshore, and islands, etc. The dataset contains four classes of targets: airplane, oilcan, bridge, and ship, consisting of 1851 bridges, 39 858 ships, 12 319 oil tanks, and 6368 airplanes.

SAR-AIRcraft-1.0 [38] is based on Gaofen-3 satellite data, with single polarization and spatial resolution of 1 m. The dataset consists of 4368 images and 16 463 aircraft targets, with image sizes of 800 × 800, 1000 × 1000, 1200 × 1200, and 1500 × 1500 in four different sizes. The images in the dataset are impact data of civil airplanes, and the specific classes include A220, A320/321, A330, ARJ21, Boeing737, Boeing787, and others in total seven classes.

Before the experiment, we need to preprocess the datasets. During the class-IL process, some of the images contain classes which are not in the same task, and for these images, we process them in such a way that we only label the classes for the current task (for example, a ship detection model is trained using ship data as the initial model, then bridges are incrementally learned based on the initial model. Well, in the initial training phase, for images that contain both ship and bridge classes, only the ship targets are labeled). In addition, MSAR is an obviously class-imbalanced dataset, and the imbalanced data affects the performance of the detector significantly, yet this is not the focus of the study of this article. Therefore, for the justification of the experiment, it is still necessary to balance the dataset before carrying out the incremental detection tests, and we take the form of randomly selecting 3000 and 950 images from the ship and oilcan images to be used for the experiments. Some examples of datasets are shown in Fig. 8.

After processing as above, the number of images and targets in each class of the two datasets are shown in Fig. 9.

In this experiment, mean average prediction (mAP) was used to measure the performance of the detector, when IoU of the predicted bounding box to the ground truth was greater than or equal to 0.5, we regarded it as correct prediction, and the detection accuracy, in this case, was noted as mAP_{50}. In subsequent experiments, detection accuracy will default to mAP_{50} unless otherwise specified.

The mAP is calculated as shown in (21), where p and r represent precision and recall percentage, respectively, n recall values are taken from low to high, and p_{i+1} represents the precision when the recall percentage is taken as r_{i+1}. K

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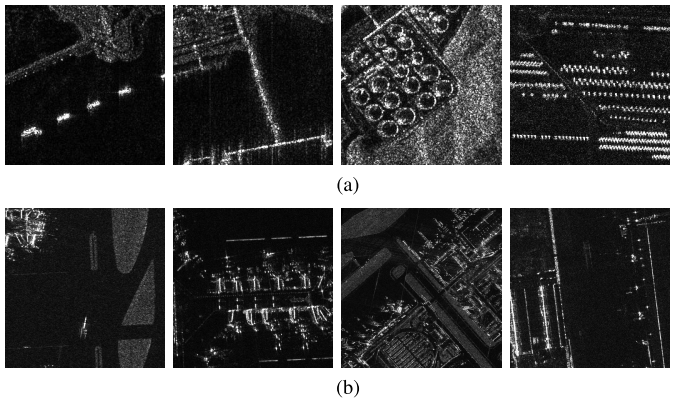


Fig. 8. Examples of (a) MSAR and (b) SARAIRcraft datasets.

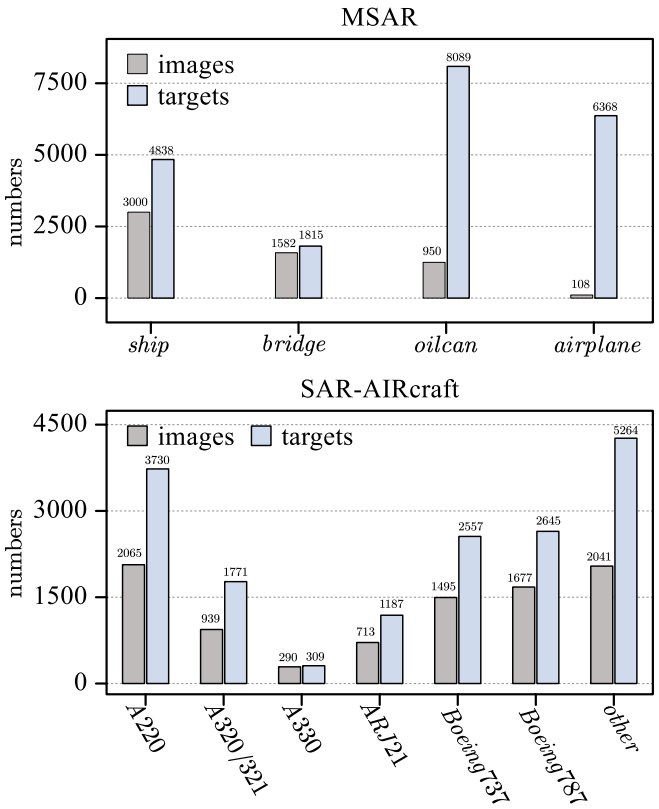


Fig. 9. Number of targets and images for each class in the MSAR and AIR-SARcraft datasets after preprocessing. The number of images here refers to the images that contain the targets in this class.

represents the number of classes, and mAP is obtained by averaging the AP for each class

$$\mathrm{AP} = \sum_{i=1}^{n-1} (r_{i+1} - r_i) \cdot p_{i+1} \ \{r_1, r_2, \ldots, r_n\}$$
$$\mathrm{mAP} = \frac{1}{K} \sum_{j=1}^{K} \mathrm{AP}_j. \tag{21}$$

B. Experimental Settings

1) Different Training Strategies: When new data comes in, we have multiple strategies to update the old model. There are four common strategies for updating: full-data (FD) training, fine-tuning (FT), CF, and class-IL.

1) *FD Training:* The model is retrained using all the old and new data to update the parameters of the whole model. The model obtained from FD training is theoretically the model with the best results, called the theoretical upper bound.
2) *FT:* Only new data is used for training, and the old branch parameters and shared layer parameters are protected by parameter isolation, and so on.
3) *CF:* The model is retrained using only new data without any adjustments, in this way the old parameters will be completely overwritten, and therefore the model will catastrophically lose the ability to detect old classes.
4) *IL:* The model is trained with new data while using old knowledge as guidance, in this way the old knowledge is protected and gets updated according to the training condition. The proposed method in this article belongs to IL, where we use the parameters of the initial model as the old knowledge.

2) Different IL Scenarios: In this article, we consider the process of data increase in practical situations, and design two kinds of incremental scenarios: one-step incremental and multistep incremental, with corresponding settings of different experimental plans.

1) *One-Step:* One-step incremental is the most basic class-incremental scenario, which means that the detection model trained by some classes as the initial model, and the initial detection model will be updated when images of new classes arrive, making the model have the ability to detect both new and old classes of targets. For this experiment, we divide the MSAR and SAR-AIRcraft datasets as follows, respectively: a) for MSAR, the number of base class is 2 and the number of new class is 2, denoted as "2+2." The base classes are ship and bridge, and the new classes are oilcan and airplane; b) for the SAR-AIRcraft dataset, the number of the base class is set to 4, and the number of the new class is set to 3, denoted as "4+3." Among them, the base class include A220, A320/321, A330, and ARJ21, and the old class include Boeing737, Boeing787, and others; and c) meanwhile, in order to verify the effect of the method presented in this article in a focused manner, we set up a cross-scene incremental scenario: the base class is ship, and the new class is oilcan.
2) *Multistep:* Considering that the increase of classes is not one-time in practical cases, this experiment is designed with a multistep incremental scenario. Typically, when new classes of data are introduced, we begin by applying the class-IL method to quickly update the model, enabling it to detect both old and new classes. Once sufficient time becomes available, a more comprehensive training process is performed to enhance the model further. As additional data is introduced, the best-performing model from the prior steps is used for pretraining, iteratively applying the update process. This multistep can be viewed as a combination of multiple one-step increments. We simulated the above process in our experiments by

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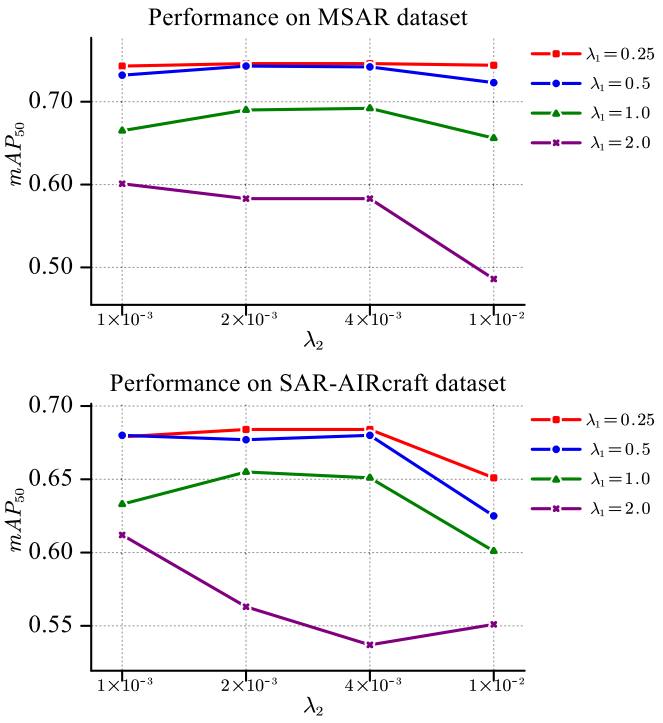


Fig. 10. Performance of our method with different parameter λ, the experimental data settings are MSAR (2+2) and SAR-AIRcraft (4+3).

dividing the dataset into multiple subdatasets, and using the subdatasets as training data and the best model from the previous stage as the pretraining model for incremental training at each stage. The specific division is as follows: a) for MSAR, the base classes were set to ship and bridge, the new class for the first time increment was oilcan, and the new class for the second time increment was the airplane, and b) for SAR-AIRcraft, the base classes are set to A220 and A320/321, the new classes for the first increment are A330 and ARJ21, the new classes for the second increment are Boeing737 and Boeing787, and the new classes for the third increment are set to other.

C. Ablation Study

1) Comparison of Performance for Different Values of λ_1 and λ_2: The total loss function (20) contains two important parameters λ_1 and λ_2 that affect the balance between the respective tasks. We perform experiments using different combinations of λ_1 and λ_2 on the MSAR and SAR-AIRcraft datasets to observe the effect of different parameter combinations on the performance. For the parameter settings, we refer to the work of [51], [52], and [54], and set the value range of λ_1 to $\{0.25, 0.5, 1.0, 2.0\}$, and the value range of λ_2 to $\{1\times10^{-3}, 2\times10^{-3}, 4\times10^{-3}, 1\times10^{-2}\}$, combining both of them, and the experimental results are shown in Fig. 10.

From the experimental results, it can be seen that the detector has the best and most stable performance when $\lambda_1 = 0.25$. When λ_1 gradually increases, the overall detection effect decreases, especially when λ_1 is larger than 0.5. For λ_2, the detection performance is best at values of 2×10^{-3} and 4×10^{-3}. In contrast, the detector performs most consistently with $\lambda_2 = 2\times10^{-3}$, giving the best results for almost all values of λ_1. Therefore, the parameter settings for all subsequent experiments were $\lambda_1 = 0.25$ and $\lambda_2 = 2\times10^{-3}$.

TABLE I
EFFECT OF EACH COMPONENT ON OUR METHOD

CAA		GBSM	Data Setting	
GR	LA		MSAR(2+2)	SAR-AIRcraft(4+3)
			72.6	60.5
✓			72.5(▼0.1)	63.4(▲2.9)
	✓		69.5(▼3.1)	61.4(▲0.9)
✓	✓		73.8(▲1.2)	67.6(▲7.1)
✓	✓	✓	74.6(▲2.0)	68.4(▲7.9)

2) Evaluation on Each Component of Our Method: The core of the proposed method consists of two components: the CAA and the GBSM. We analyze the effect of each component in a one-step dataset division manner. In addition, the CA module consists of the GR module and LA, and we also add them as independent parts of the experiment. The performance on the MSAR (2+2) and SAR-AIRcraft (4+3) datasets is shown in Table I.

From the results, it can be seen that CA and GBSM lead to an improvement in the overall performance of the incremental detector, with a 2% and 7.9% improvement on both datasets when they work together, as well as a positive effect on the detector when working alone. However, it is noteworthy that on the MSAR (2+2), the GR and the LA on their own resulted in a decrease in the performance instead, but the combination of the two parts together to form the CA module resulted in an increase as well. We suggest that GR and LA are concerned with global and local information, respectively, using them individually will make the method pay too much attention to a certain part of information and thus cause performance decrease, and when working together they have a balancing effect on each other.

D. Comparison With Other Methods

In the following, we verify the effectiveness of the proposed method through comparison experiments. Our method is implemented based on the GFL [31] detection network, and in order to see more obviously the performance of different methods under the same condition, we choose SID [32] and ERD [30] as the comparison methods, which are both implemented based on the single-stage detector.

We validate the performance of the proposed method in three aspects, in addition to the previously described one-step and multistep incremental scenarios, we similarly observed the time consumption of the proposed method experimentally to validate the efficiency advantage of the class-IL methods.

1) One-Step: At first, we performed comparison experiments on MSAR (2+2) and SAR-AIRcraft (4+3) datasets. The overall results are shown in TableII, where the red numbers represent the best performance and the Diff represents the difference from the FD detection results. Meanwhile,

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TABLE II

EXPERIMENTAL RESULTS OF DIFFERENT METHODS FOR ONE-STEP INCREMENTAL SETUP

Data Setting	Method	Base Classes		New Classes		All Classes	
		mAP	Diff	mAP	Diff	mAP	Diff
MSAR 2+2	Ori	81.7	-	-	-	-	-
	Full-Data	79.8	-	79.2	-	79.5	-
	SID	76.4	-3.4	44.9	-34.3	61.7	-17.8
	ERD	75.4	-4.4	69.8	-9.4	72.6	-6.9
	Ours(+CAA)	**75.6**	**-4.2**	**71.9**	**-7.3**	**73.8**	**-5.7**
	Ours(+CAA+GBSM)	**76.1**	**-3.7**	**73.2**	**-6.0**	**74.6**	**-4.9**
SAR-AIR 4+3	Ori	75.7	-	-	-	-	-
	Full-Data	78.8	-	74.1	-	76.8	-
	SID	76.2	-2.6	55.3	-18.8	67.3	-9.5
	ERD	74.6	-4.2	41.6	-32.5	60.5	-16.3
	Ours(+CAA)	**75.9**	**-2.9**	**56.5**	**-17.6**	**67.6**	**-9.2**
	Ours(+CAA+GBSM)	**76.0**	**-2.8**	**58.3**	**-15.8**	**68.4**	**-8.4**

TABLE III

EXPERIMENTAL RESULTS FOR EACH CLASS OF DIFFERENT METHODS UNDER ONE-STEP INCREMENTAL SETUP (MSAR DATASET)

Method	Old Classes		New Classes		Average
	ship	bridge	oilcan	airplane	
Ori	79.5	84.0	-	-	-
Full-Data	79.1	80.6	97.4	60.9	79.5
SID	73.1	79.7	89.8	0.04	61.7
ERD	73.6	77.1	96.0	43.6	72.6
Ours(+CAA)	**74.1**	**77.1**	**96.9**	**46.9**	**73.8**
Ours(+CAA+GBSM)	**74.6**	**77.6**	**97.3**	**49.0**	**74.6**

we recorded the effect of our method when only the CA module is used, which is noted as Ours(+CAA) in the table, while Ours(+CAA+GBSM) represents the method that uses all the components of this article. From the experimental results, it can be seen that our method is the best performance method on both datasets. Ours(+CAA) and Ours(+CAA+GBSM) show higher mAP than the other methods on entire data, with the least performance penalty compared to FD training. Notably, our method slightly underperforms SID in the detection of base classes, by 0.3% and 0.2% on MSAR (2+2) and SAR-AIRcraft (4+3) datasets, respectively, but far better than SID in the detection of new classes, exceeding the performance by 28.3% and 3.0%.

The detailed detection results are shown in Tables III and IV, Table III shows the AP for each class in MSAR (2+2) dataset and Table IV shows the AP for each class in SAR-AIRcraft (4+3) dataset. From Table III, Ours(+CAA+GBSM) gets the highest AP in the ship, oilcan, and airplane, trails SID in the bridge, and outperforms all other methods in overall mAP. From Table IV, it can be seen that Ours(+CAA+GBSM) achieves the best results in four classes: A220, A320/321, Boeing787, and other, especially the AP of Boeing787 is much higher than that of other methods, and Ours(+CAA) achieves the best results in ARJ21 and Boeing737, except for the AP of A330 which is lower than that of SID. From the overall results, it seems that our method has the best performance in the one-step incremental scenario.

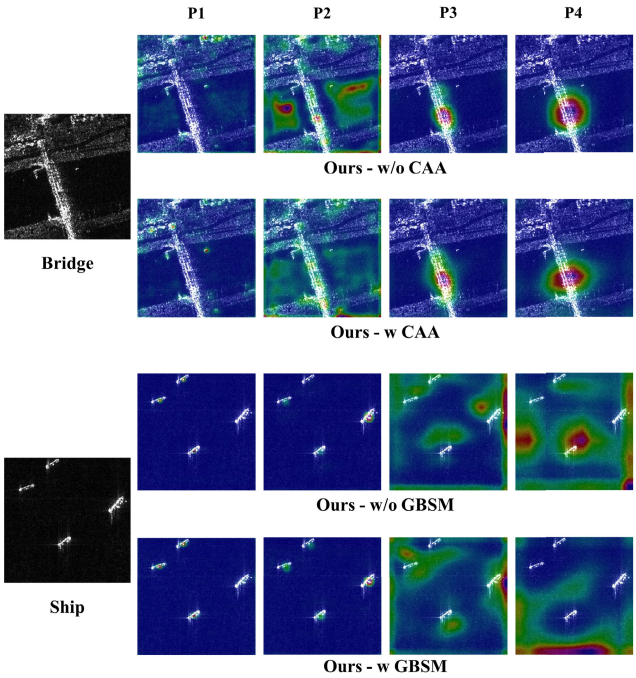


Fig. 11. Feature heat maps from different models. p1 to p4 represent the outputs from different layers of the FPN.

Finally, we performed experiments in cross-scene incremental scenarios. We used the ship as the base class and the oilcan as the new class, and the background of the ship image is usually a pure black sea surface, while the background of the oilcan image is land. The experimental results are shown in Table V, similarly we recorded the AP metrics for a variety of conditions. As can be seen from the results, our method shows the best performance on all metrics and is very close to FD training on the most commonly detected metrics, AP_{50} and $AP_{50:95}$, with only 0.6% and 0.8% performance penalty.

In addition, we observed the model feature maps in the class-incremental experiments. As shown in Fig. 11, we selected the mapping of the bridge and ship feature maps on the original image in cross-scene increment, and the feature maps were chosen to be the output of the FPN at layers p1 p4. Since the bridge and ship images also have some scene differences, we compared the feature maps with/without the CAA (w CAA and w/o CAA) module and with/without the GBSM (w GBSM and w/o GBSM). As can be seen from the feature maps of Fig. 11-bridge, after using the CAA module, the attention of the feature at p2 spreads out globally instead of focusing attention on the bridge itself; the attention on the bridge itself is more accurate at layer p3. As can be seen from the feature maps in Fig. 11-ship, after using the GBSM, the attention at layer p2 is clearer, especially for the target at the uppermost part of the image, which is completely lost in the feature maps without GBSM, while the feature map with GBSM still notices the target. The effectiveness of the method is further proved by the view of the attention map.

2) Multistep: Table VI shows the multistep class-incremental detection results on the MSAR dataset with two steps of class addition. The base classes are ship and bridge, the new class learned in the first increment

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TABLE IV
EXPERIMENTAL RESULTS FOR EACH CLASS OF DIFFERENT METHODS UNDER ONE-STEP INCREMENTAL SETUP (SAR-AIRCRAFT DATASET)

Method	Old Classes				New Classes			Average
	A220	A320/321	A330	ARJ21	Boeing737	Boeing787	other	
Ori	63.9	86.0	78.0	74.8	-	-	-	-
Full-Data	66.9	89.8	77.2	81.2	65.4	77.4	79.4	76.8
SID	60.2	88.0	80.2	76.5	50.8	54.5	60.6	67.3
ERD	58.1	87.3	78.4	74.5	39.0	35.4	50.5	60.5
Ours(+CAA)	**60.2**	**89.0**	**77.2**	**77.1**	**51.6**	**56.1**	**61.9**	**67.6**
Ours(+CAA+GBSM)	**60.9**	**89.2**	**77.2**	**76.5**	**51.0**	**60.1**	**63.8**	**68.4**

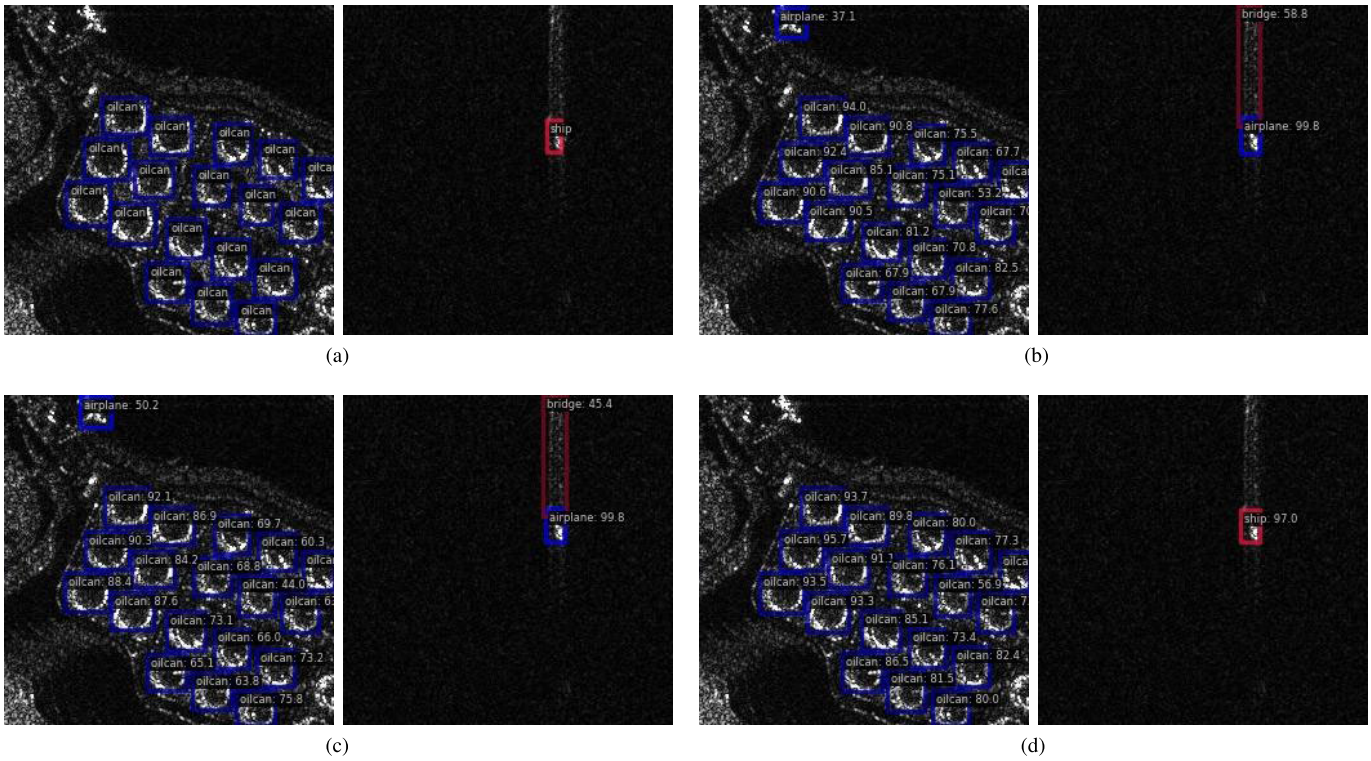


Fig. 12. Experimental results on the MSAR dataset. The experimental setting was multistep incremental and the resultant figures are from step 2. (a) Ground truth. (b) SID. (c) ERD. (d) Ours.

TABLE V
CROSS-SCENE INCREMENTAL SCENARIO WITH INITIAL CLASS AS SHIP AND NEW CLASS AS OILCAN

Method	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	$AP_{50:95}$
Full-Data	61.4	39.9	40.4	43.5	62.7	42.2
SID	39.0	18.0	19.1	22.8	20.5	19.8
ERD	58.7	38.3	39.2	42.2	40.0	39.7
Ours	**60.8**	**40.1**	**40.7**	**43.9**	**44.1**	**41.4**

is oilcan, and at the second increment, the new class is airplane. From the results, it can be seen that at step 1, our method is better than the comparison methods in terms of performance, with Ours(+CAA) being the method with the highest mAP for both the new classes and overall, and Ours(+CAA+GBSM) achieved the highest mAP on the old classes. However, it is worth noting that in step 1, the overall detection performance of Ours(+CAA+GBSM) is slightly lower than that of Ours(+CAA), which suggests that GBSM plays a negative role in this incremental step. In step 2, Ours(+CAA+GBSM) achieved the best results on both old and new classes, with the highest detection rates for bridge, oilcan, and airplane, except for the AP on ship class, which was 0.1% lower than that of Ours(+CAA).

Table VII shows the multistep class-incremental detection results on the SAR-AIRcraft dataset, due to the more classes in this dataset, we designed a three times class-incremental process. The base classes are A220 and A320/321, the new classes for the first increment are A330 and ARJ21, for the second increment, are Boeing737 and Boeing787, and for the third increment is other. From the detection results, it can be seen that in step 1, our method is the best-performing method, and Ours(+CAA+GBSM) achieves the highest mAP on both the

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TABLE VI
EXPERIMENTAL RESULTS OF DIFFERENT METHODS FOR MULTISTEP INCREMENTAL ON MSAR-1.0 DATASET

Phase	Method	mAP (old/new/all)	Classes				Diff (old/new/all)
			ship	bridge	oilcan	airplane	
Initial	Full-Data(class 1-2)	-/-/81.7	79.5	84.0	-	-	-
Step 1 (class2 +1)	Full-Data(class 1-3)	82.2/40.7/68.4	80.5	83.9	40.7	-	-
	SID	67.5/15.7/50.2	75.1	59.8	15.7	-	-17.4/-25.0/-18.2
	ERD	73.9/28.6/58.7	77.7	70.0	28.6	-	-8.3/-12.1/-9.7
	Ours(+CAA)	**74.1/33.6/60.6**	**77.8**	**70.4**	**33.6**	**-**	**-8.1/-7.1/-7.8**
	Ours(+CAA+GBSM)	**76.5/26.0/59.6**	**76.7**	**76.3**	**26.0**	**-**	**-5.7/-14.7/-8.8**
Step 2 (class3 +1)	Full-Data(class 1-4)	85.7/60.9/79.5	79.1	80.6	97.4	60.9	-
	SID	57.5/46.0/54.6	76.7	81.0	14.8	46.0	-28.2/-14.9/-24.9
	ERD	67.1/47.4/62.2	76.5	80.8	44.0	47.4	-18.6/-13.5/-17.3
	Ours(+CAA)	**71.3/51.3/66.3**	**76.9**	**80.7**	**56.2**	**51.3**	**-14.4/-9.6/-13.2**
	Ours(+CAA+GBSM)	**73.2/53.0/68.2**	**76.8**	**81.3**	**61.5**	**53.0**	**-12.5/-7.9/-11.3**

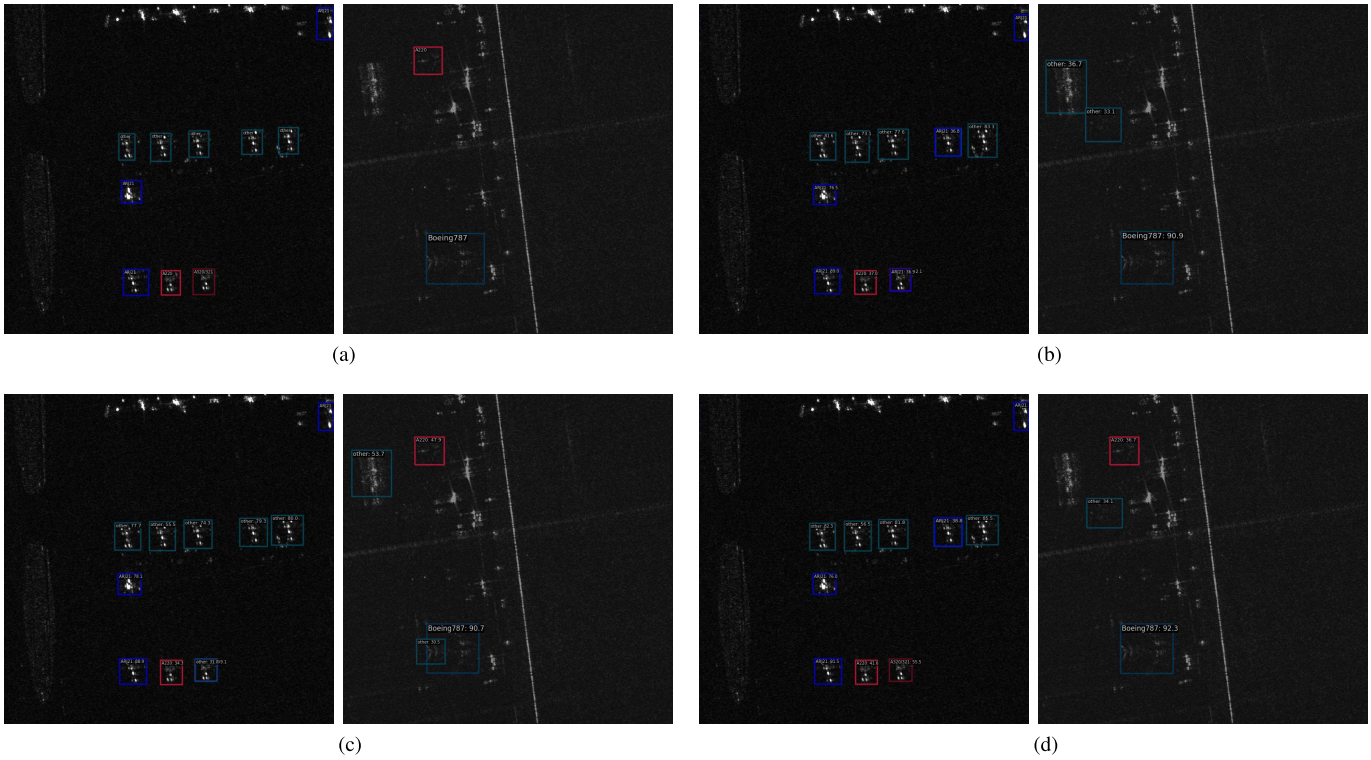


Fig. 13. Experimental results on the SAR-AIRcraft dataset. The experimental setting was multistep incremental and the resultant figures are from step 3. (a) Ground truth. (b) SID. (c) ERD. (d) Ours.

new classes and old classes. In step 2, our method is still the best-performing method overall, where Ours(+CAA+GBSM) slightly performs better than Ours(+CAA), with an overall mAP 0.2% higher. At step 3, our method is still the best-performing method, but all incremental detection methods perform well at this phase, with Ours(+CAA+GBSM) outperforming the worst-performing method, SID, by 1.5% in terms of mAP.

We show some experimental results, as shown in Figs. 12 and 13. Fig. 12 is from the MSAR dataset and Fig. 13 is from the SAR-AIRcraft dataset, both of which are the last phase in a multistep incremental scenario. The results show that our method is overall better than the other incremental detection methods. However, there are still some false alarms and missed detectable comparing to the ground truth.

3) Efficiency: We evaluate the efficiency by comparing the time consumption of the training process under both class-IL and FD training. We chose the multistep incremental tests on the SAR-AIRcraft dataset as the observation object to record the time consumption at each phase, which was chosen because it has the most experimental steps. The same learning conditions are used for each training session to ensure the fairness of the experiment. We compare the efficiency of the methods in two aspects, one is the total time consumption with the same number of training epochs, and the other is the time consumption when reaching the optimal performance on each phase.

Following the above idea, we first observed the total time consumption. There are three incremental steps in total, and we set the same 24 epochs for each phase to record the time of

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TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS FOR MULTISTEP INCREMENTAL ON SAR-AIRCRAFT-1.0 DATASET

Phase	Method	mAP (old/new/all)	Classes							Diff (old/new/all)
			A220	A320/321	A330	ARJ21	Boeing737	Boeing787	other	
Initial	Full-Data(class 1-2)	-/-/70.2	62.3	78.2	-	-	-	-	-	-
Step 1 (class2 +2)	Full-Data(class 1-4)	74.9/76.4/75.7	63.9	86.0	78.0	74.8	-	-	-	-
	SID	67.9/68.0/68.0	54.9	81.0	77.0	59.3	-	-	-	-7.0/-8.4/-7.7
	ERD	68.8/61.7/65.2	56.6	80.9	61.4	62.0	-	-	-	-6.1/-14.7/-10.5
	Ours(+CAA)	**68.8/63.3/66.1**	**62.5**	**75.1**	**63.8**	**62.8**	**-**	**-**	**-**	**-6.1/-13.1/-9.1**
	Ours(+CAA+GBSM)	**70.1/71.7/70.9**	**62.8**	**77.4**	**75.8**	**67.5**	**-**	**-**	**-**	**-3.8/-4.7/-4.8**
Step 2 (class4 +2)	Full-Data(class 1-6)	74.3/71.0/73.2	57.8	93.9	77.2	68.4	64.1	77.8	-	-
	SID	73.0/57.9/68.0	53.9	86.9	79.5	71.8	52.6	63.3	-	-1.3/-13.1/-5.2
	ERD	74.5/56.0/68.3	56.8	87.5	78.0	75.6	53.1	58.9	-	+0.2/-15.0/-4.9
	Ours(+CAA)	**74.5/59.1/69.3**	**57.4**	**87.6**	**77.2**	**75.8**	**54.0**	**64.1**	**-**	**-0.4/-11.9/-3.9**
	Ours(+CAA+GBSM)	**73.6/61.4/69.5**	**53.8**	**87.7**	**77.2**	**75.7**	**58.1**	**64.6**	**-**	**-0.7/-9.6/-3.7**
Step 3 (class6 +1)	Full-Data(class 1-7)	76.3/79.4/76.8	66.9	89.8	77.2	81.2	65.4	77.4	79.4	-
	SID	73.3/70.0/72.8	57.1	90.9	77.2	73.9	62.6	78.2	70.0	-3.0/-9.4/-4.0
	ERD	74.1/66.0/73.0	59.2	94.5	77.2	72.3	64.3	77.8	66.0	-2.2/-13.4/-3.8
	Ours(+CAA)	**74.9/68.1/73.9**	**60.5**	**93.7**	**77.2**	**72.9**	**65.3**	**79.3**	**68.1**	**-1.4/-11.3/-2.9**
	Ours(+CAA+GBSM)	**75.0/70.5/74.3**	**61.2**	**94.6**	**77.0**	**72.6**	**65.1**	**79.1**	**70.5**	**-1.3/-8.9/-2.5**

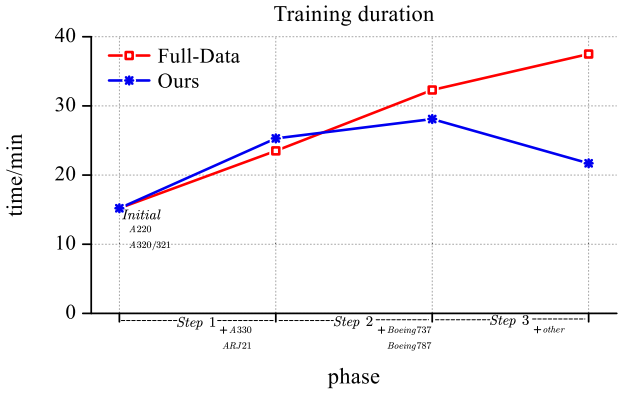


Fig. 14. Time consumption records of multistep incremental experiments on the SAR-AIRcraft dataset. As can be seen from the figure, the efficiency advantage of the IL method becomes more and more obvious as the incremental stages increase.

our method and FD training. The results are shown in Fig. 14, where we can see that in the step 1 phase, our method time consumption is slightly higher than FD training. After that, the time consumption of our method is lower than FD training, and the efficiency advantage becomes more and more obvious as the phase increases. The experimental results are in line with our expectation, since the incremental detection method needs to be applied to KD, the processing time of a single image will be higher than the general training. However, the IL method has the advantage of using only new data, so the total time consumption of IL is gradually less than FD training as the incremental steps increase. It is predictable that the efficiency advantage of IL will be more significant for larger data incremental processes.

Furthermore, we observed the time required for the model to reach the optimal value (the highest mAP, denoted as "Top") and the optimal region (the interval where performance stabilizes within 1% of the optimal value, denoted as "Top 1%"). It is well known that different methods have different speeds of loss convergence during training as well as different time consumption to find the optimal solution. Usually, if the method is efficient in convergence, then a smaller number of training sessions can be set at the beginning of training. Therefore, we similarly set the training epoch to 24, and record the validation results of each epoch detection model. The validation results at each phase are shown in Fig. 15. The lines in the figure are the validation results of the model trained in the current epoch, the first row is the epoch that reaches the highest mAP for the first time, and the second row is the optimal region of the interval. From Fig. 15, we can see that our method reaches the highest mAP ahead of the FD training, and reaches the optimal region more quickly, which is due to the fact that our method keeps the old knowledge. The time consumption to reach the optimal value and the optimal region is shown in Fig. 16, which shows that, combining with the convergence speed of the algorithm, the efficiency advantage of the IL method is more obvious, and the time consumption in each phase is lower than FD training. At the same time, the efficiency advantage is gradually enlarged with the increase of the incremental phases.

V. DISCUSSION

In all the comparison experiments, our method consistently achieved the best performance. However, the experimental results also reveal some shortcomings in our method.

First, in the multistep incremental experiment using the MSAR dataset, we found that GBSM played a negative role in the first step increment. From the experimental results, it can be seen that the overall mAP of Ours(+CAA+GBSM) is slightly lower than that of Ours(+CAA), although the detection is still best on the old class. Through observation, we found that oilcans are arranged very densely in images, and in such cases, precise boundary regression is crucial for

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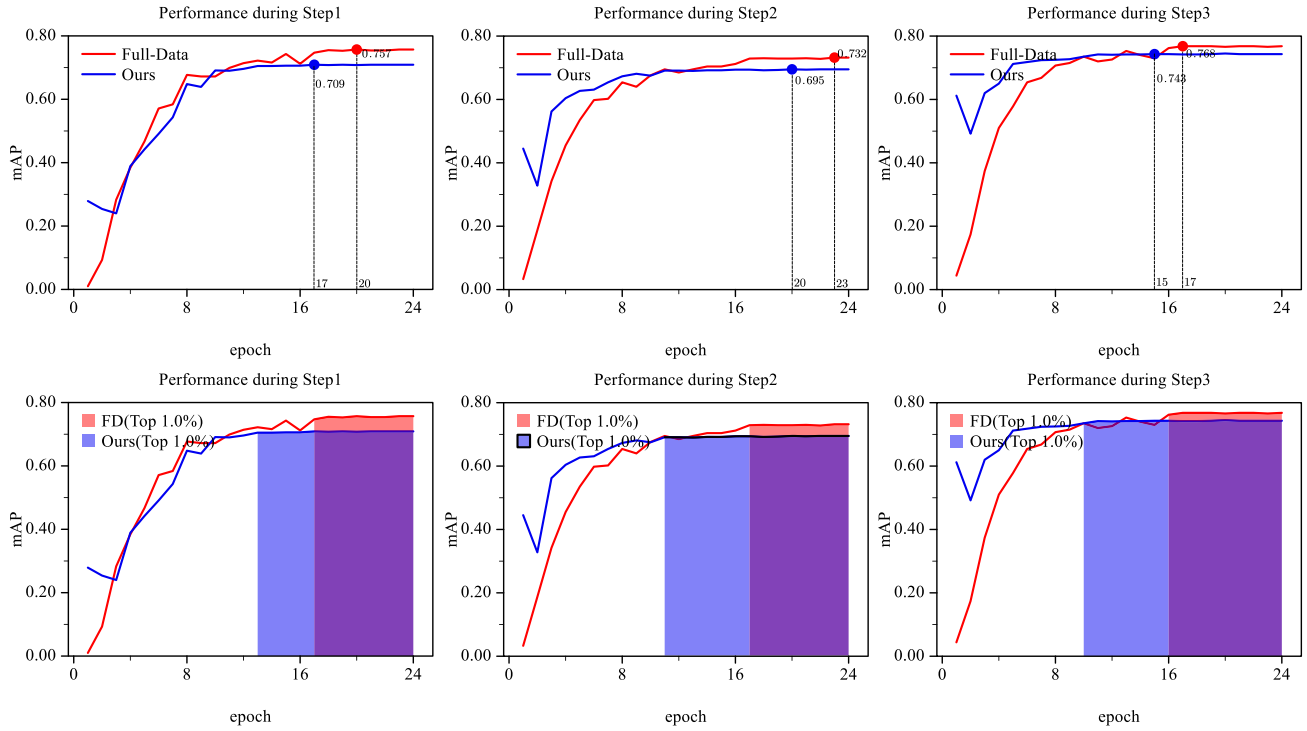


Fig. 15. Model performance and time consumption during the experiments. FD represents the FD training and Ours represents the ITD method proposed in this article. From the figure, it can be seen that the IL method always reaches the optimal performance interval before the FD training strategy. Combined with the time consumption in Fig. 14, that can be concluded that the IL method has a great advantage in efficiency.

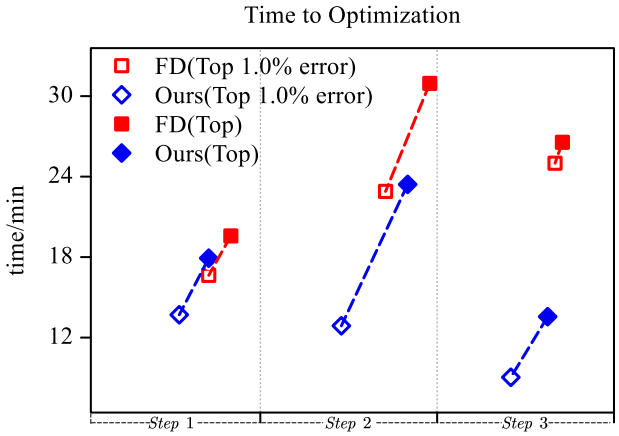


Fig. 16. Time consumption to achieve optimal detection performance on each data incremental step. The ITD shows efficiency advantages at all phases due to the fact that it uses less training data and the losses converge faster.

accurate localization. However, since GBSM reduces boundary sensitivity, it may have a negative effect in this situation. This requires us to make further improvements.

Second, we discovered that the ITD method resulted in an increase in the false alarm rate. The reason for this phenomenon is that in some SAR images containing multiple classes of targets, certain classes are not labeled during the IL process. This can lead to an increase in false alarms as the detector may incorrectly recognize unlabelled classes in these images. This problem is unavoidable, both for experiments and real applications. Some potential targets in the images are also discovered gradually and labeled later. However, this incompletely labeled data tends to confuse the detector's judgment. Some research has already paid attention to this problem, mainly focusing on the field of open-environment target detection [56], [57], which is likewise an issue we need to explore further.

Finally, our method is based on KD implementation, which is the same as the current mainstream ITD methods. As stated in Section II, there are other ways that ITD can be implemented. Meta-learning, in particular, has now achieved good results in the areas of few-shot target detection [12]. If meta-learning can be combined with ITD methods, it can further lead to success in few-shot ITD, which needs to be further investigated in the future.

VI. Conclusion

To address the issue of reduced robustness in SAR incremental target detectors caused by scene changes, this article proposes the CAA and GBSM to enhance the adaptability of SAR incremental target detectors. The CAA plays a critical role in the feature knowledge transfer stage of incremental training. By converting feature maps into global relation maps and LA maps, it effectively captures the contextual relationship, thereby strengthening the model's perception of the interactions between targets and backgrounds. The GBSM method focuses on optimizing incremental localization loss. By constructing 2-D Gaussian distribution models of target bounding boxes and measuring the differences between these distributions, it achieves effective suppression of the sensitivity of target boundaries in the localization loss of incremental detectors, while increasing the model's focus on the central displacement and overall distribution differences

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of targets. The incremental detection scheme combining CAA and GBSM not only enables efficient model updates without reliance on original data but also specifically optimizes for the characteristics of SAR imagery, significantly improving the stability and adaptability of the detector when faced with scene changes.

Comparison experiments on the MSAR and SAR-AIRcraft datasets show that our method achieves the best detection accuracy in both one-step incremental and multistep incremental scenarios. The localization accuracy in the scenario of cross-scene is even better than other methods. Meanwhile, in this article, the training time of IL is recorded, and the efficiency of the IL method is proved to be more advantageous by comparing it with the FD training. But it is worth noting that although our method achieves better results, there are still some shortcomings in. For example, GBSM may limit the detector's ability to learn new knowledge in some scenarios, as well as cause a few false alarms, which need to be further analysis and research.

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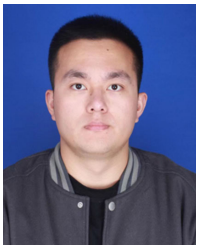
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